DEVOPS

LAB 03

Working with Git History, Stashing,
and Reverting Commits

Name: Fatima Noor Ul Imran

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Section: A

GitHub Username: fatimanoorulimran

Repository Links

Submission repository: https://github.com/fatimanoorulimran/DevOps

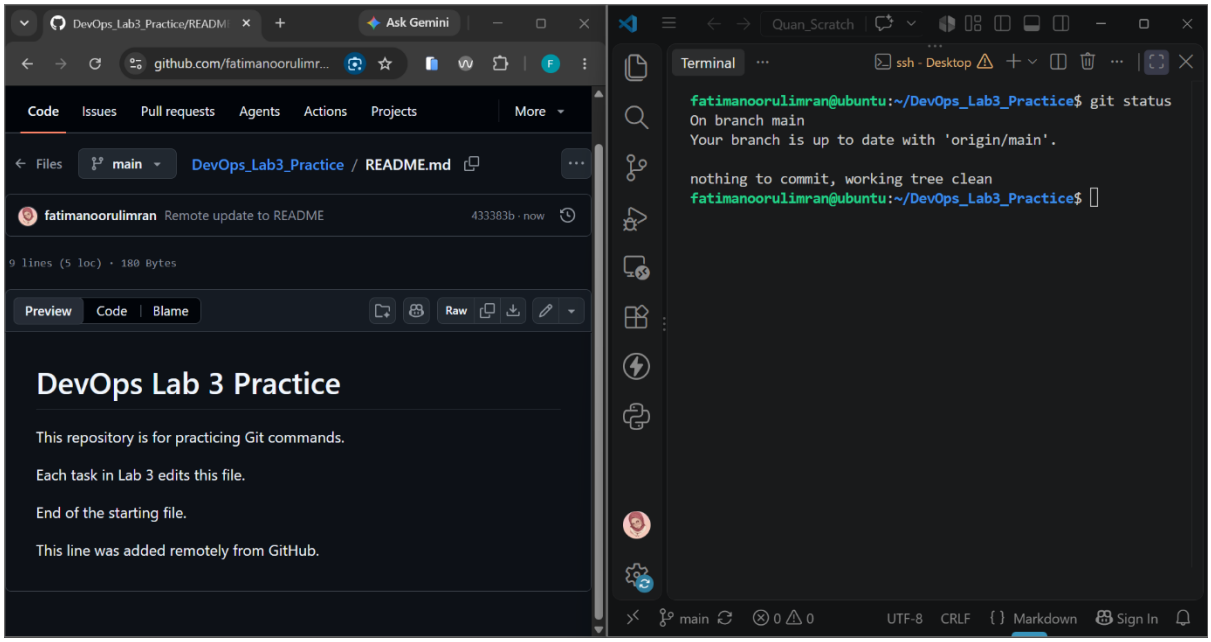
Practice repository: https://github.com/fatimanoorulimran/DevOps_Lab3_Practice

Gitea fork: https://github.com/fatimanoorulimran/Gitea

Pages repository: https://github.com/fatimanoorulimran/fatimanoorulimran.github.io

Live site: https://fatimanoorulimran.github.io

Task 1: Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)



Line added to README.md directly on GitHub and committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ sed -i '4a This line was added locally.\n' README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps Lab 3 Practice

This repository is for practicing Git commands.

This line was added locally.

Each task in Lab 3 edits this file.

End of the starting file.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Local update to README"
[main 4da1930] Local update to README
 1 file changed, 2 insertions(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Different line added to README.md locally and committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:fatimanoorulimran/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Push rejected because the remote contains commits I do not have locally.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git pull --no-rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 969 bytes | 64.00 KiB/s, done.
From github.com:fatimanoorulimran/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   330edd2..433383b  main       -> origin/main
Auto-merging README.md
Merge made by the 'ort' strategy.
 README.md | 2 ++
 1 file changed, 2 insertions(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

git pull --no-rebase merged the remote and local commits, creating a merge commit.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 618 bytes | 154.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   433383b..f2f9797  main -> main
branch 'main' set up to track 'origin/main'.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Push succeeded after the merge.

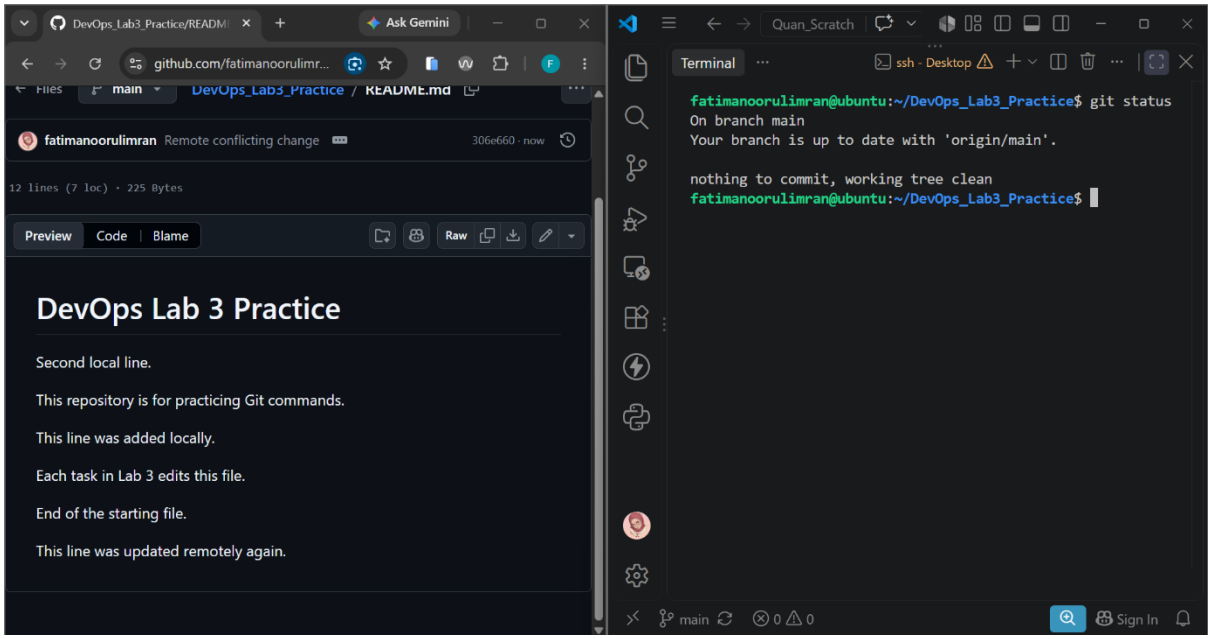
```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 971 bytes | 138.00 KiB/s, done.
From github.com:fatimanoorulimran/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   076c2ff..87f7b00  main       -> origin/main
Successfully rebased and updated refs/heads/main.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

git pull --rebase replayed my local commit on top of the remote commits.

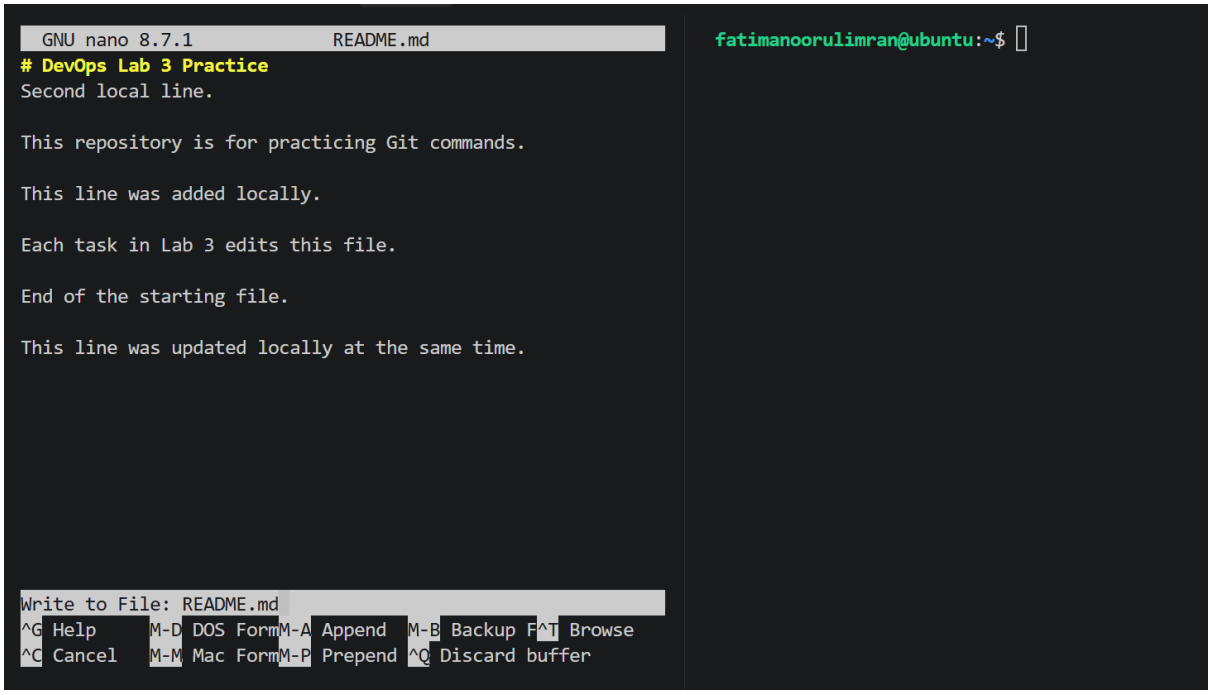
```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 311 bytes | 155.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   87f7b00..6c78846  main -> main
branch 'main' set up to track 'origin/main'.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Push succeeded after the rebase, with a linear history.

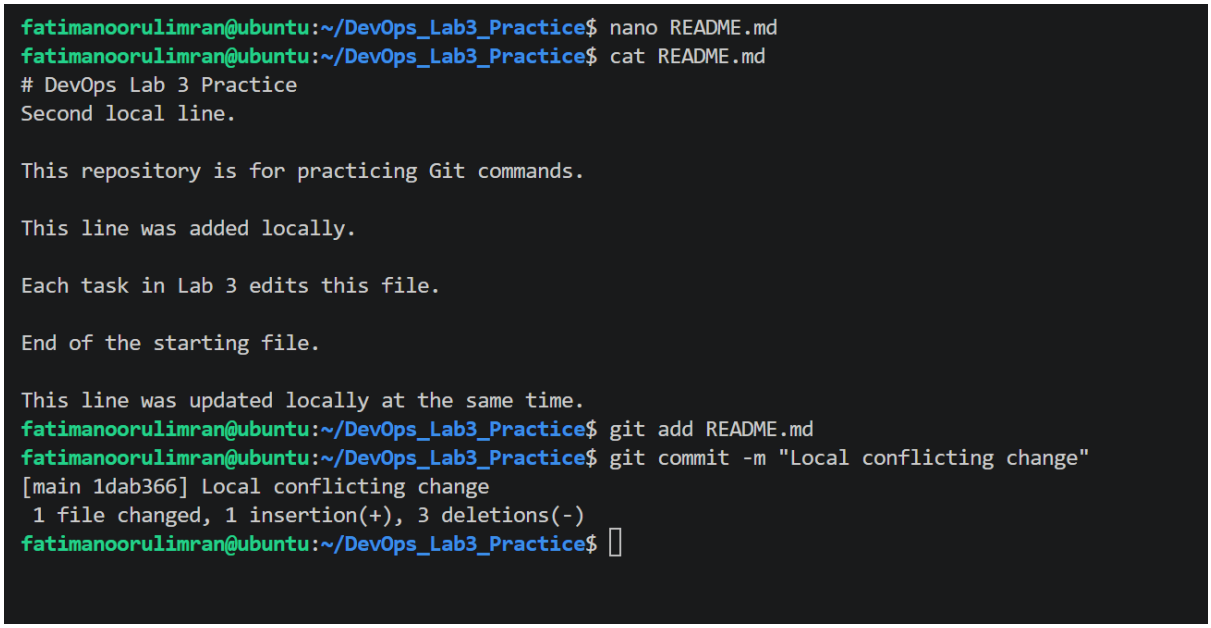
Task 2: Creating and Resolving Merge Conflicts Manually



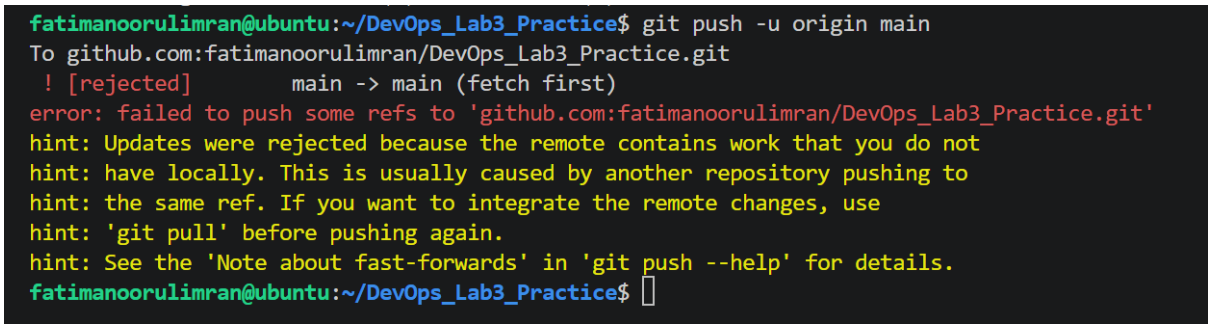
Line changed on GitHub and committed as Remote conflicting change.



The same line changed differently on my local machine.



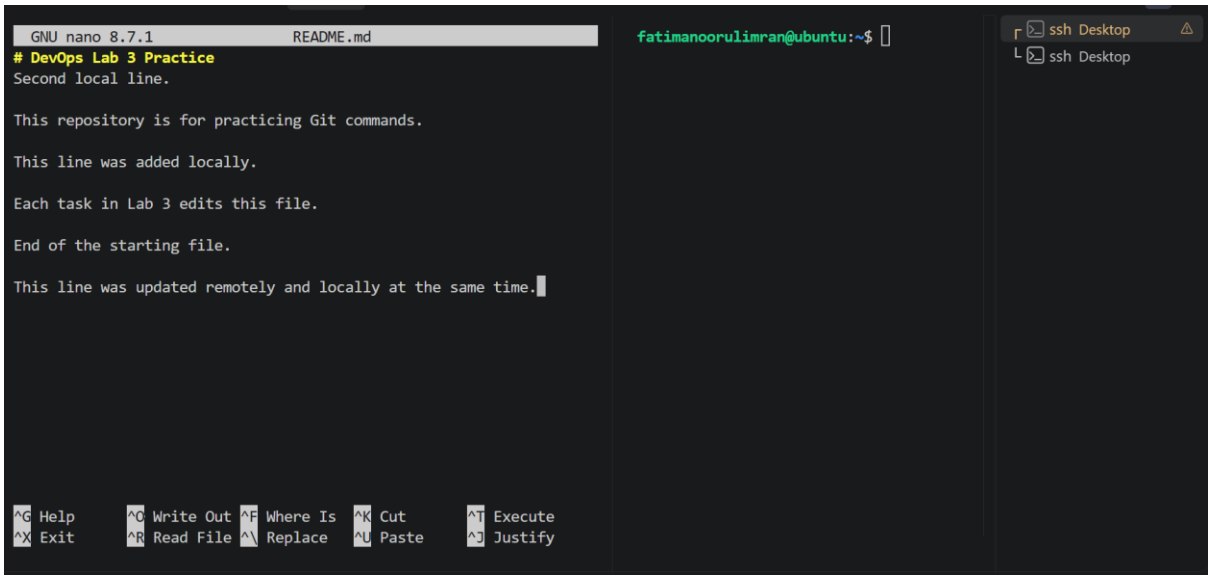
Local conflicting change committed.



Push rejected because the remote has a conflicting change.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 983 bytes | 327.00 KiB/s, done.
From github.com:fatimanoorulimran/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   6c78846..306e660  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply 1dab366... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 1dab366... # Local conflicting change
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

git pull --rebase paused and reported a conflict in README.md.



Conflict markers removed and the file edited by hand to keep both changes.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git rebase --continue
[detached HEAD 5bd44d3] Local conflicting change
 1 file changed, 1 insertion(+), 1 deletion(-)
Successfully rebased and updated refs/heads/main.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

File staged with git add and the rebase finished with git rebase --continue.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 326 bytes | 163.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   306e660..5bd44d3  main -> main
branch 'main' set up to track 'origin/main'.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Resolved changes pushed to GitHub.

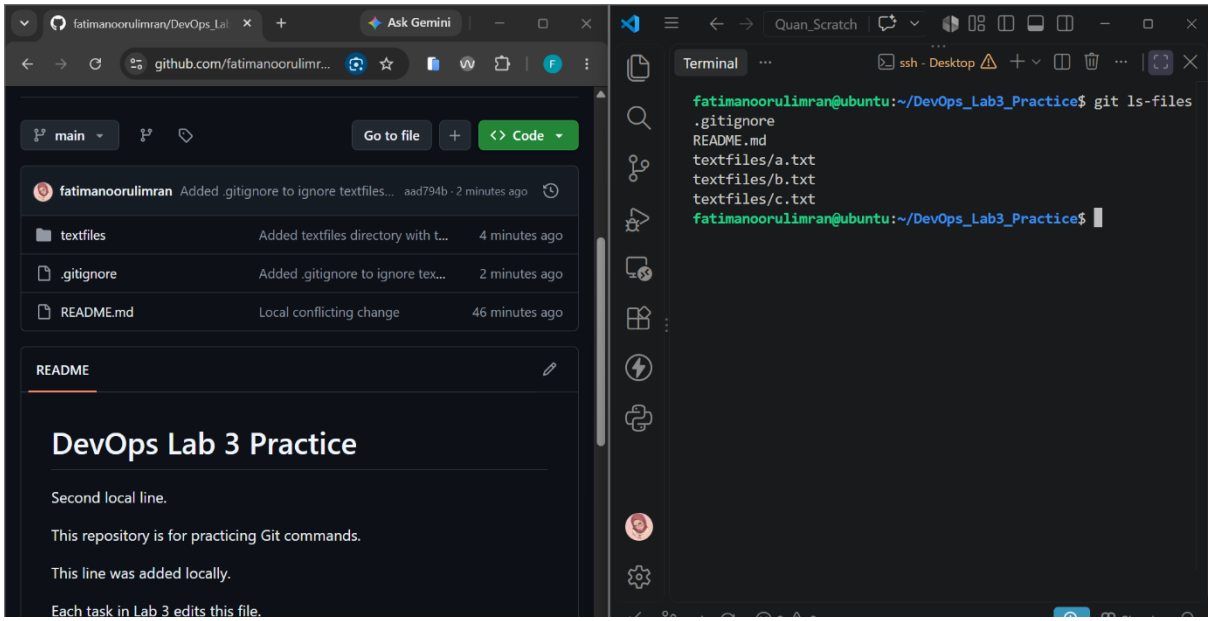
Task 3: Managing Ignored Files with .gitignore and Removing Tracked Files

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ mkdir textfiles
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "File A content" > textfiles/a.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "File B content" > textfiles/b.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "File C content" > textfiles/c.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add .
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added textfiles directory with three files"
[main 66d71f7] Added textfiles directory with three files
 3 files changed, 3 insertions(+)
 create mode 100644 textfiles/a.txt
 create mode 100644 textfiles/b.txt
 create mode 100644 textfiles/c.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 469 bytes | 156.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   5bd44d3..66d71f7  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

textfiles folder with three files committed and pushed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "textfiles/*" > .gitignore
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add .gitignore
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added .gitignore to ignore textfiles directory"
[main aad794b] Added .gitignore to ignore textfiles directory
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 358 bytes | 119.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   66d71f7..aad794b  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

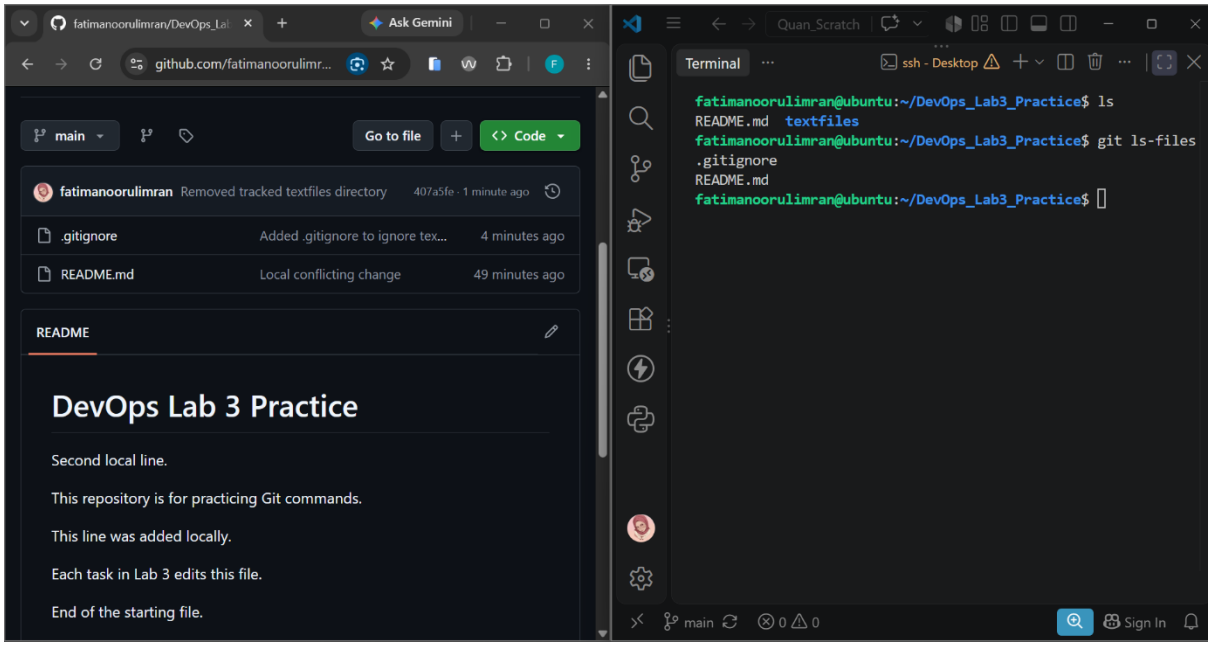
.gitignore created to ignore textfiles/* and pushed.



GitHub still shows textfiles because the files were already tracked.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add .
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Removed tracked textfiles directory"
[main 407a5fe] Removed tracked textfiles directory
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 249 bytes | 249.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   aad794b..407a5fe  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Files untracked with git rm -r --cached, then committed and pushed.



textfiles folder removed from GitHub but still present locally.

Task 4: Create Temporary Changes and Use git stash

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Feature test line one" >> README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Feature test line two" >> README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Feature test line three" >> README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Changes the README.md file"
[feature-branch 72750e7] Changes the README.md file
 1 file changed, 3 insertions(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

feature-branch created and README.md changes committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Uncommitted line one" >> README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Uncommitted line two" >> README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Switching to main failed because of uncommitted changes.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git stash
Saved working directory and index state WIP on feature-branch: 72750e7 Changes the README.md file
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git stash list
stash@{0}: WIP on feature-branch: 72750e7 Changes the README.md file
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Uncommitted changes saved with git stash.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git branch
  feature-branch
* main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Switched to main successfully after stashing.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git checkout feature-branch
Switched to branch 'feature-branch'
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git branch
* feature-branch
  main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Returned to feature-branch.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch feature-branch
nothing to commit, working tree clean
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Working tree clean because the changes are in the stash.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (31424c2570442475d6961798f32a90591127052f)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Stashed changes restored with git stash pop.

Task 5: Checkout a Specific Commit Using git log

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
407a5fe (HEAD -> main, origin/main, origin/HEAD) Removed tracked textfiles directory
aad794b Added .gitignore to ignore textfiles directory
66d71f7 Added textfiles directory with three files
5bd44d3 Local conflicting change
306e660 Remote conflicting change
6c78846 Second local update
87f7b00 Second commit again remotely
076c2ff Local commit
eaec52e Second commit again remotely
c70c5b4 Second remote update
f2f9797 Merge branch 'main' of github.com:fatimanoorulimran/DevOps_Lab3_Practice
4da1930 Local update to README
433383b Remote update to README
330edd2 Restore README body
27421bf Remote update to README
4c961db Initial README
84b50b8 Initial README
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Commit history shown with git log --oneline.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git checkout 076c2ff
Note: switching to '076c2ff'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 076c2ff Local commit
```

Older commit checked out, putting Git in detached HEAD state.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Previous HEAD position was 076c2ff Local commit
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Returned to the main branch.

Task 6: Resetting Commits (Soft vs Hard Reset)

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Test line one" >> README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add .
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added test line"
[main fb5905c] Added test line
 1 file changed, 1 insertion(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

First test line committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Test line two" >> README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add .
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Second test commit"
[main 5db26b4] Second test commit
 1 file changed, 1 insertion(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Second test line committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
5db26b4 (HEAD -> main) Second test commit
fb5905c Added test line
407a5fe (origin/main, origin/HEAD) Removed tracked textfiles directory
aad794b Added .gitignore to ignore textfiles directory
66d71f7 Added textfiles directory with three files
5bd44d3 Local conflicting change
306e660 Remote conflicting change
6c78846 Second local update
87f7b00 Second commit again remotely
076c2ff Local commit
eaec52e Second commit again remotely
c70c5b4 Second remote update
f2f9797 Merge branch 'main' of github.com:fatimanoorulimran/DevOps_Lab3_Practice
4da1930 Local update to README
433383b Remote update to README
330edd2 Restore README body
27421bf Remote update to README
4c961db Initial README
84b50b8 Initial README
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

History showing both commits before the reset.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git reset --soft HEAD~1
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md

fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

git reset --soft HEAD~1 undid the last commit.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
fb5905c (HEAD -> main) Added test line
407a5fe (origin/main, origin/HEAD) Removed tracked textfiles directory
aad794b Added .gitignore to ignore textfiles directory
66d71f7 Added textfiles directory with three files
5bd44d3 Local conflicting change
306e660 Remote conflicting change
6c78846 Second local update
87f7b00 Second commit again remotely
076c2ff Local commit
eaec52e Second commit again remotely
c70c5b4 Second remote update
f2f9797 Merge branch 'main' of github.com:fatimanoorulimran/DevOps_Lab3_Practice
4da1930 Local update to README
433383b Remote update to README
330edd2 Restore README body
27421bf Remote update to README
4c961db Initial README
84b50b8 Initial README
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

History after the soft reset, with the second commit gone.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps Lab 3 Practice
Second local line.

This repository is for practicing Git commands.

This line was added locally.

Each task in Lab 3 edits this file.

End of the starting file.

This line was updated remotely and locally at the same time.
Test line one
Test line two
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Both edits still present in the file after the soft reset.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at fb5905c Added test line
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

git reset --hard HEAD~1 removed the last commit and its changes.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
fb5905c (HEAD -> main) Added test line
407a5fe (origin/main, origin/HEAD) Removed tracked textfiles directory
aad794b Added .gitignore to ignore textfiles directory
66d71f7 Added textfiles directory with three files
5bd44d3 Local conflicting change
306e660 Remote conflicting change
6c78846 Second local update
87f7b00 Second commit again remotely
076c2ff Local commit
eaec52e Second commit again remotely
c70c5b4 Second remote update
f2f9797 Merge branch 'main' of github.com:fatimanoorulimran/DevOps_Lab3_Practice
4da1930 Local update to README
433383b Remote update to README
330edd2 Restore README body
27421bf Remote update to README
4c961db Initial README
84b50b8 Initial README
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

History after the hard reset.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps Lab 3 Practice
Second local line.

This repository is for practicing Git commands.

This line was added locally.

Each task in Lab 3 edits this file.

End of the starting file.

This line was updated remotely and locally at the same time.
Test line one
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Second edit gone from the file after the hard reset.

Task 7: Amending the Last Commit

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Amend test line" >> README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add .
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Fix log message"
[main 6d46a57] Fix log message
 1 file changed, 1 insertion(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Commit created with the message Fix log message.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Amend test notes" >> notes.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add .
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit --amend
[main c6c86cb] Fix log message and add notes
 Date: Sun Oct 18 06:25:15 2026 +0000
 2 files changed, 2 insertions(+)
 create mode 100644 notes.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Second file added to the same commit with git commit --amend.

Task 8: Reverting a Commit (Safe Undo on Remote Branch)

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "temporary text" >> temp.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add .
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added temporary text file"
[main 233cc66] Added temporary text file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 931 bytes | 93.00 KiB/s, done.
Total 10 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   407a5fe..233cc66  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

temp.txt committed and pushed to main.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
233cc66 (HEAD -> main, origin/main, origin/HEAD) Added temporary text file
c6c86cb Fix log message and add notes
fb5905c Added test line
407a5fe Removed tracked textfiles directory
aad794b Added .gitignore to ignore textfiles directory
66d71f7 Added textfiles directory with three files
5bd44d3 Local conflicting change
306e660 Remote conflicting change
6c78846 Second local update
87f7b00 Second commit again remotely
076c2ff Local commit
eaec52e Second commit again remotely
c70c5b4 Second remote update
f2f9797 Merge branch 'main' of github.com:fatimanoorulimran/DevOps_Lab3_Practice
4da1930 Local update to README
433383b Remote update to README
330edd2 Restore README body
27421bf Remote update to README
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git revert 233cc66
[main 077a542] Revert "Added temporary text file"
 1 file changed, 1 deletion(-)
 delete mode 100644 temp.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Commit reverted with git revert, creating a new undo commit.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 270 bytes | 135.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   233cc66..077a542  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Revert commit pushed to GitHub.

Task 9: Force Push (With Caution)

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b test-force
Switched to a new branch 'test-force'
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git branch
  feature-branch
  main
* test-force
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Branch test-force created.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "force push test" > force.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add force.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Add force test file"
[test-force f06fd05] Add force test file
 1 file changed, 1 insertion(+)
 create mode 100644 force.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Small change committed on test-force.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 374 bytes | 187.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/fatimanoorulimran/DevOps_Lab3_Practice/pull/new/test-force
remote:
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
 * [new branch]      test-force -> test-force
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

test-force pushed to GitHub.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at 077a542 Revert "Added temporary text file"
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Local branch reset back one commit with git reset --hard HEAD~1.

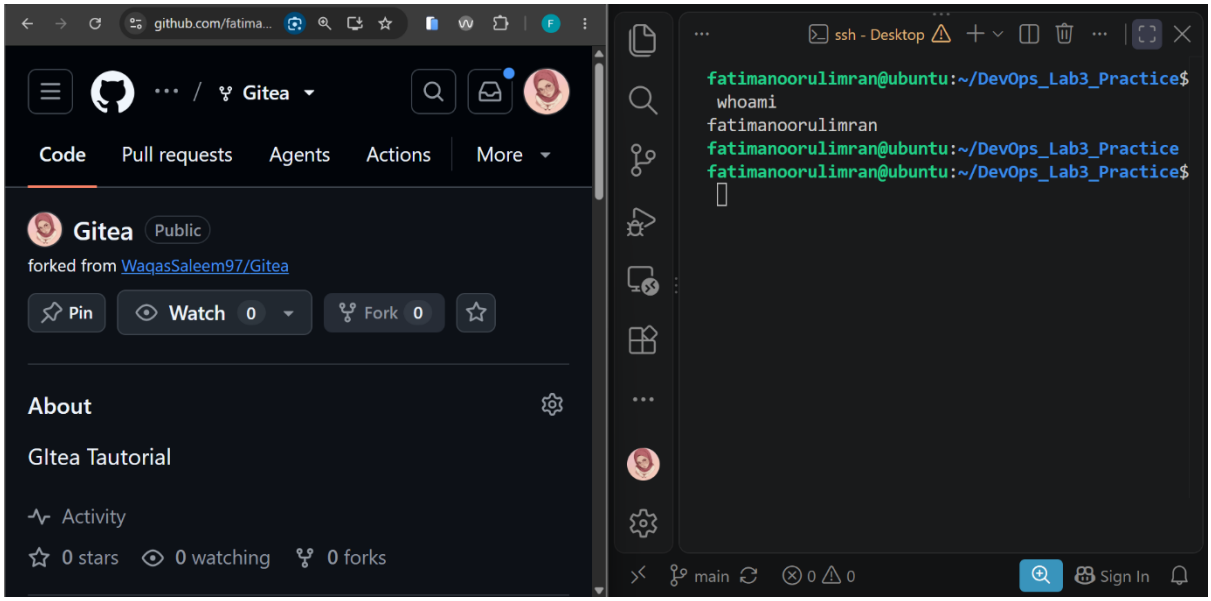
```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:fatimanoorulimran/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Normal push rejected because the remote has a commit my branch no longer has.

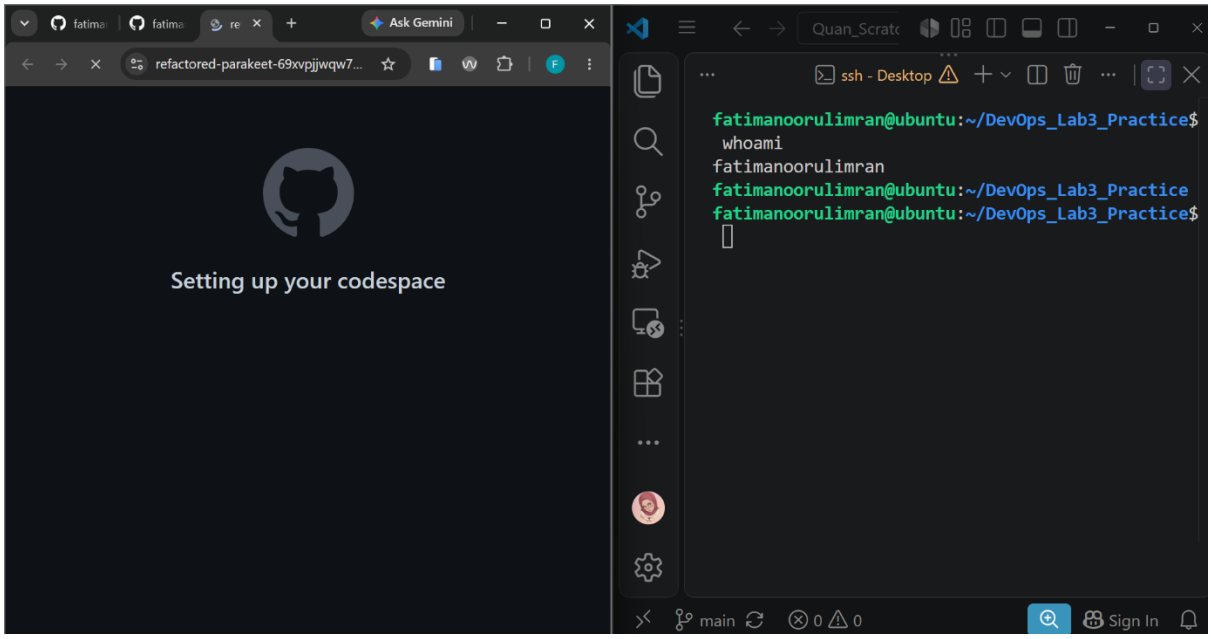
```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
 + f06fd05...077a542 test-force -> test-force (forced update)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

git push --force overwrote the remote branch history.

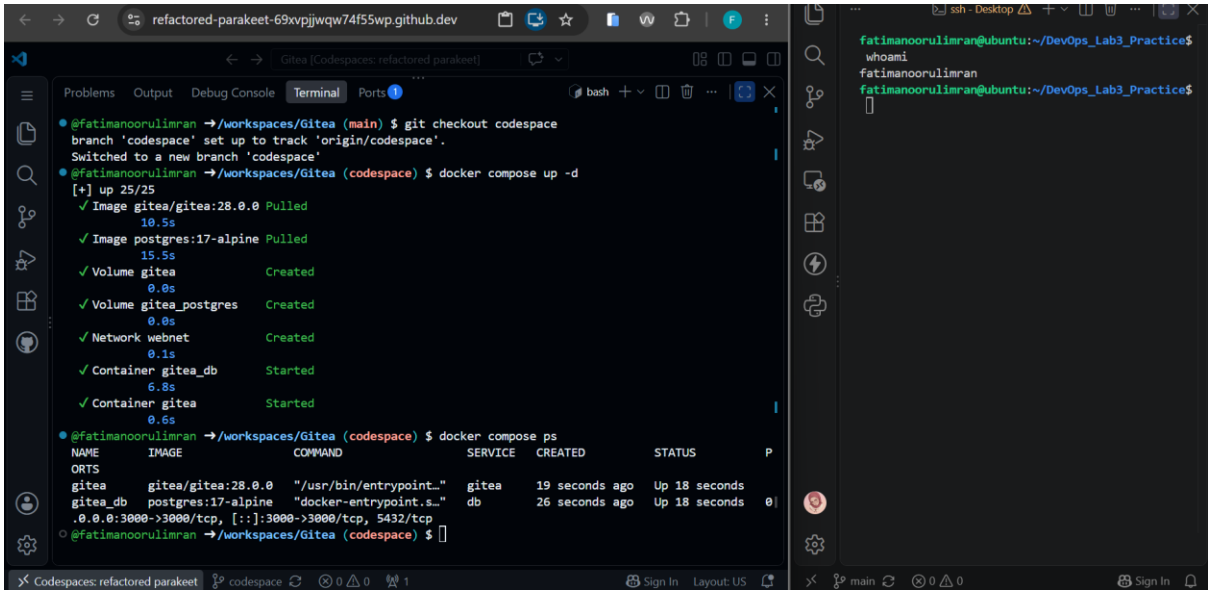
Task 10: Running Gitea in GitHub Codespaces via Docker Compose



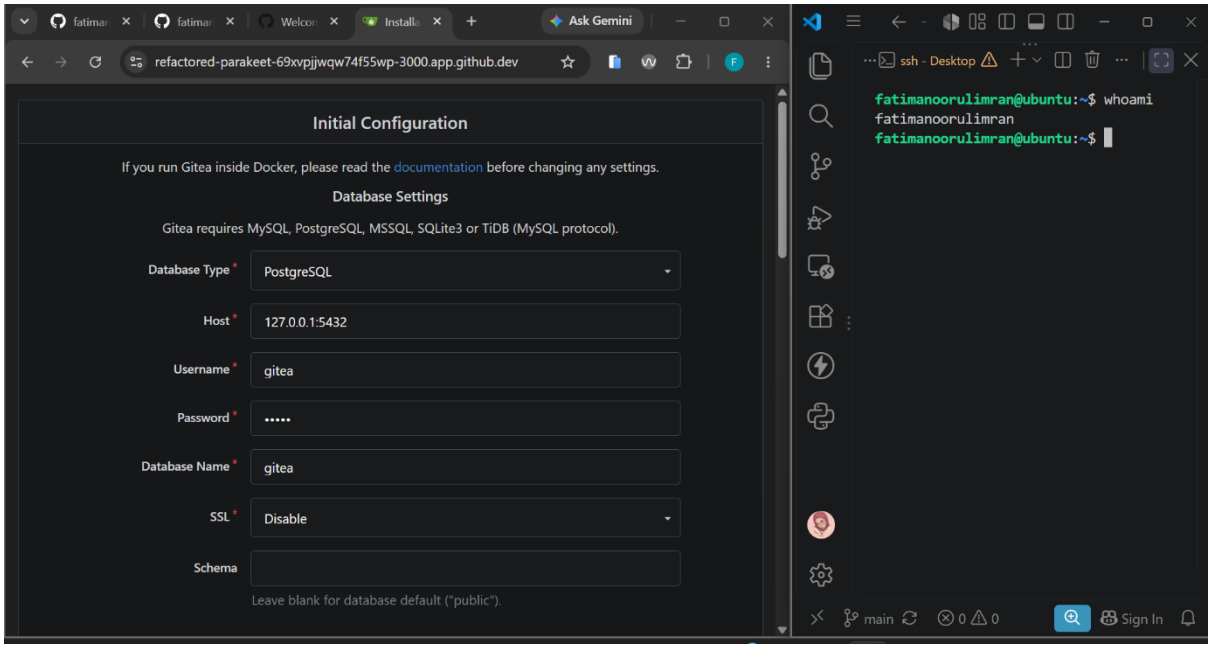
Gitea repository forked into my GitHub account.



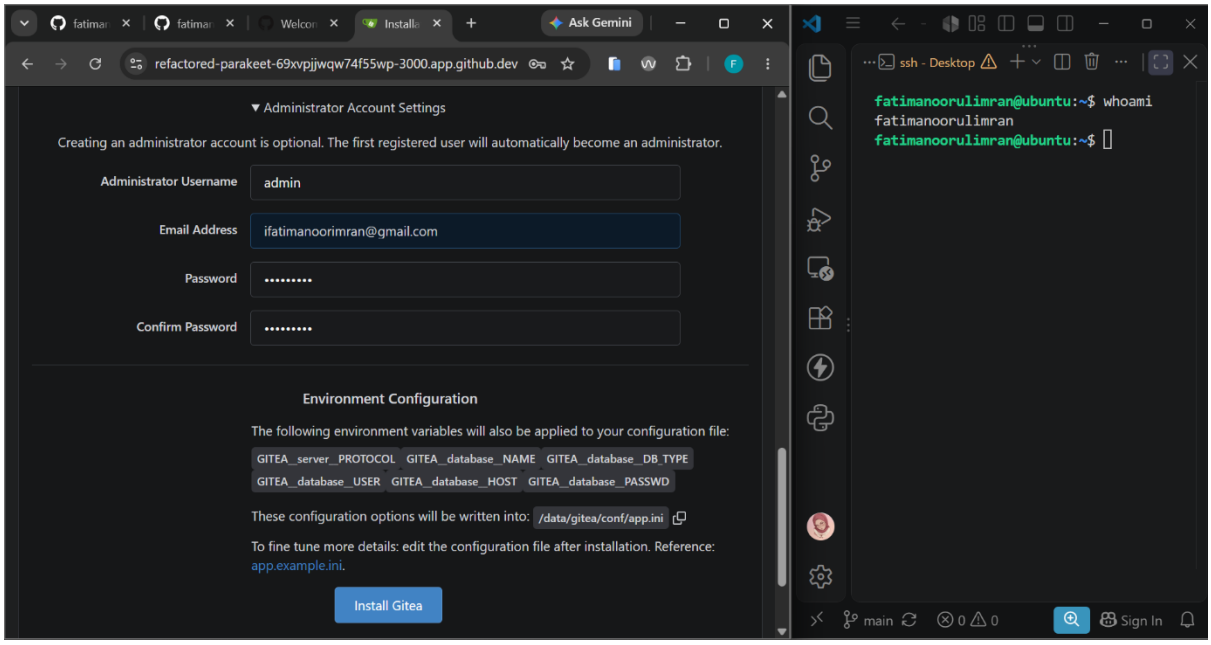
Codespace being created from the fork.



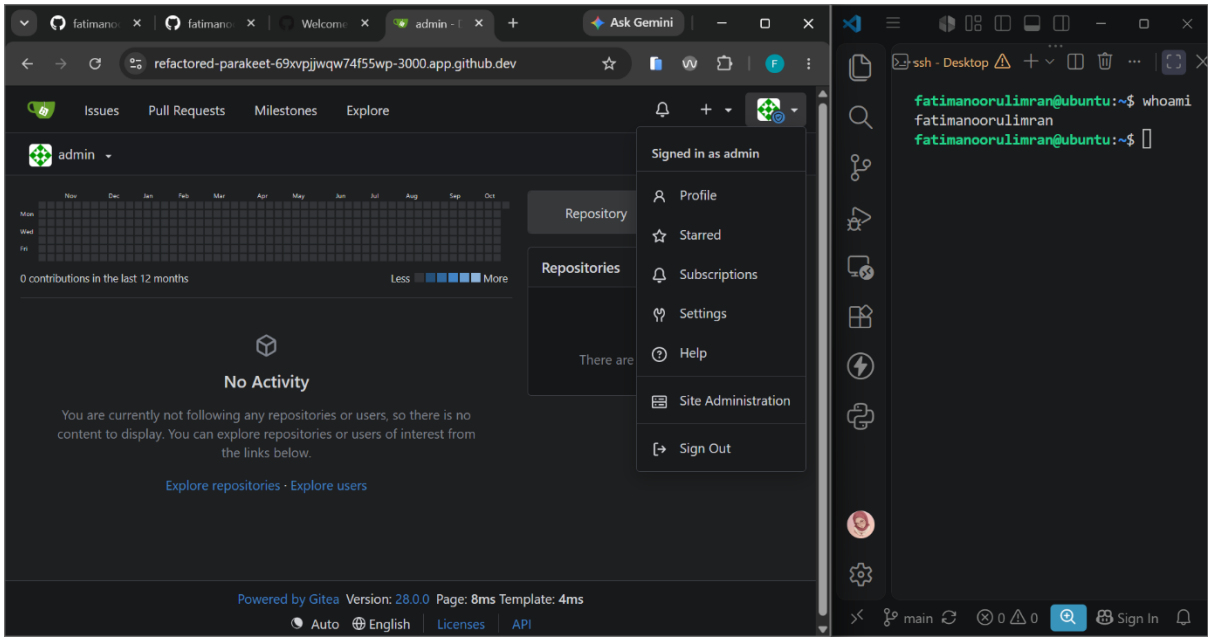
docker compose up -d started the Gitea containers.



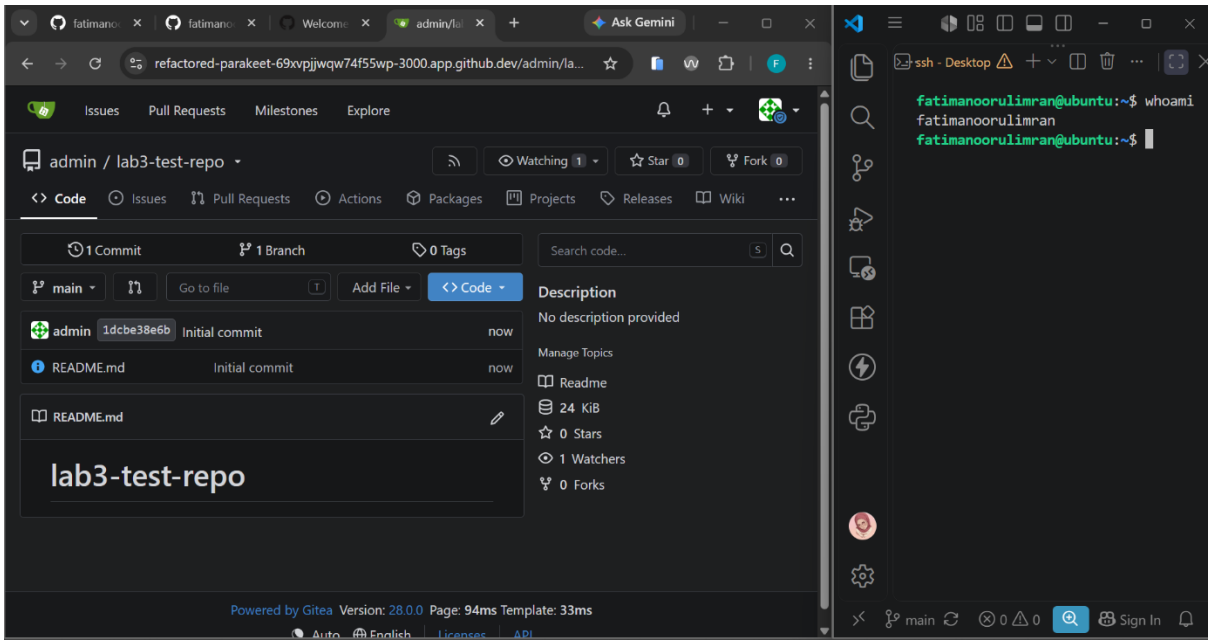
Gitea installation page opened through the forwarded public port 3000.



Administrator account configured in the installer (password not shown).

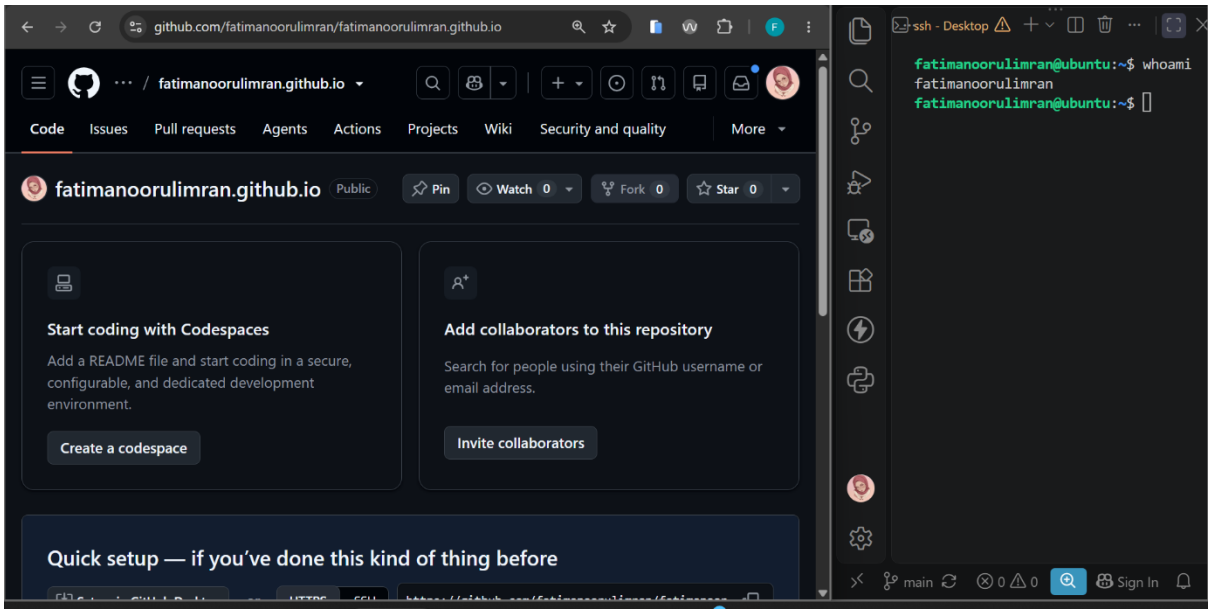


Gitea dashboard after logging in.



New repository created in Gitea.

Task 11: Creating a GitHub Pages Portfolio Site



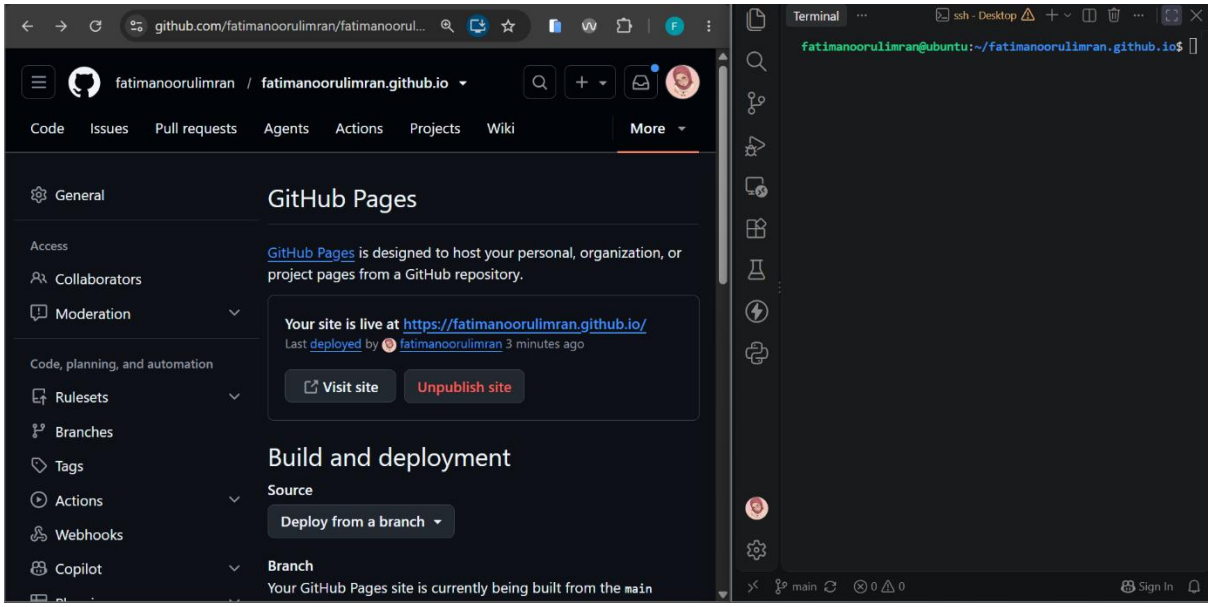
Public repository fatimanoorulimran.github.io created.

```
fatimanoorulimran@ubuntu:~$ cd ~/fatimanoorulimran.github.io
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$ nano index.html
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$ grep -rn "Iqra" .
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$ pwd
/home/fatimanoorulimran/fatimanoorulimran.github.io
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$ ls -la
total 28
drwxrwxr-x 2 fatimanoorulimran fatimanoorulimran 4096 Oct 18 08:57 .
drwxr-x--- 9 fatimanoorulimran fatimanoorulimran 4096 Oct 18 07:05 ..
-rw-rw-r-- 1 fatimanoorulimran fatimanoorulimran 5465 Oct 18 07:05 images.png
-rw-rw-r-- 1 fatimanoorulimran fatimanoorulimran 4480 Oct 18 08:57 index.html
-rw-rw-r-- 1 fatimanoorulimran fatimanoorulimran 1779 Oct 18 07:05 style.css
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$
```

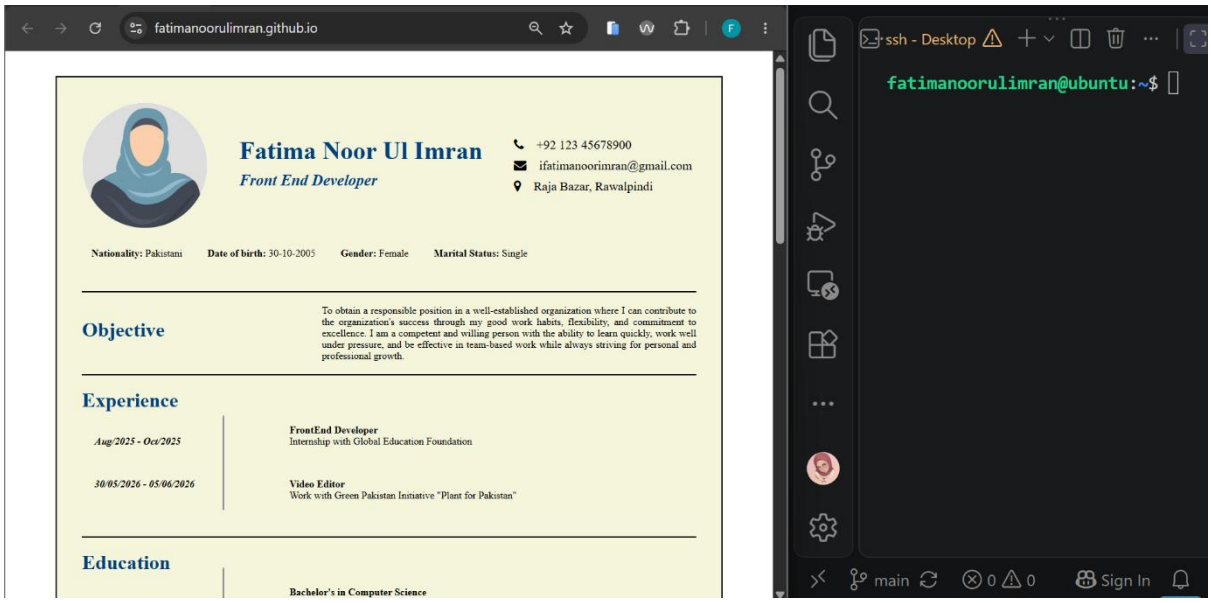
Local folder containing the static website files, including index.html.

```
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$ git add .
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$ git commit -m "Add portfolio site for GitHub Pages"
[master (root-commit) 9f85772] Add portfolio site for GitHub Pages
 3 files changed, 306 insertions(+)
 create mode 100644 images.png
 create mode 100644 index.html
 create mode 100644 style.css
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$ git branch -M main
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$ git remote add origin git@github.com:fatimanoorulimran/fatimanoorulimran.github
.io.git
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 7.93 KiB | 902.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:fatimanoorulimran/fatimanoorulimran.github.io.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
fatimanoorulimran@ubuntu:~/fatimanoorulimran.github.io$
```

Website files committed and pushed to the main branch.

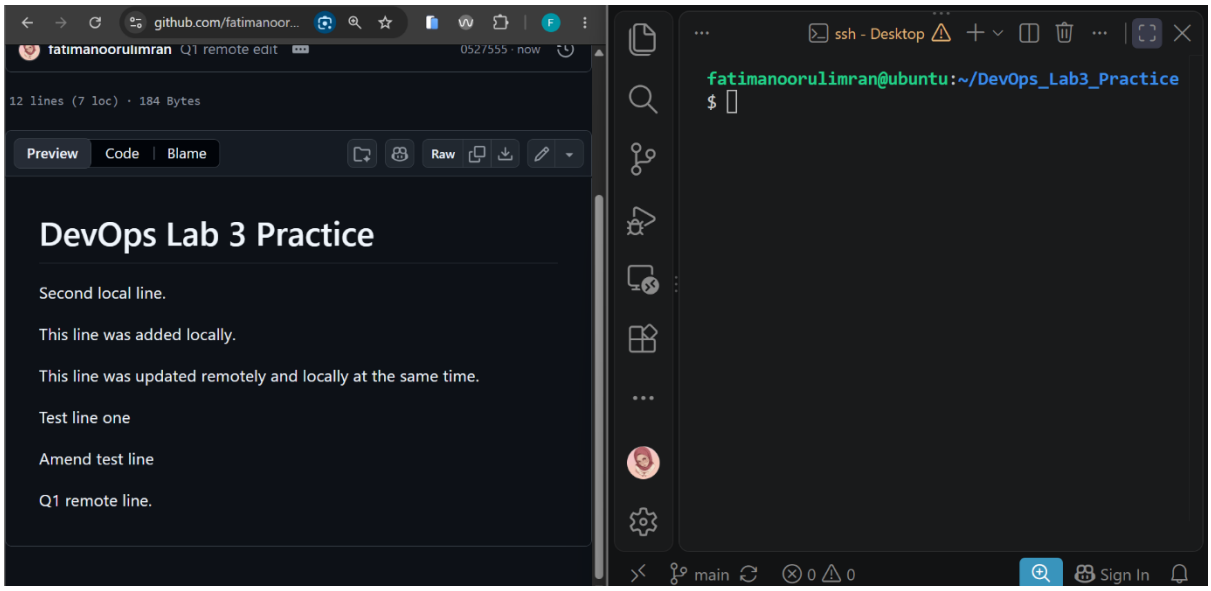


Settings > Pages showing the site published from the main branch.



Live portfolio site opened in the browser.

Exam Question 1: Local vs Remote Conflict Resolution



File edited and committed on GitHub.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ sed -i '2i Q1 local line.' README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Q1 local edit"
[main acfebb1] Q1 local edit
 1 file changed, 1 insertion(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Same file edited differently locally and committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:fatimanoorulimran/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Push rejected because the remote is ahead.

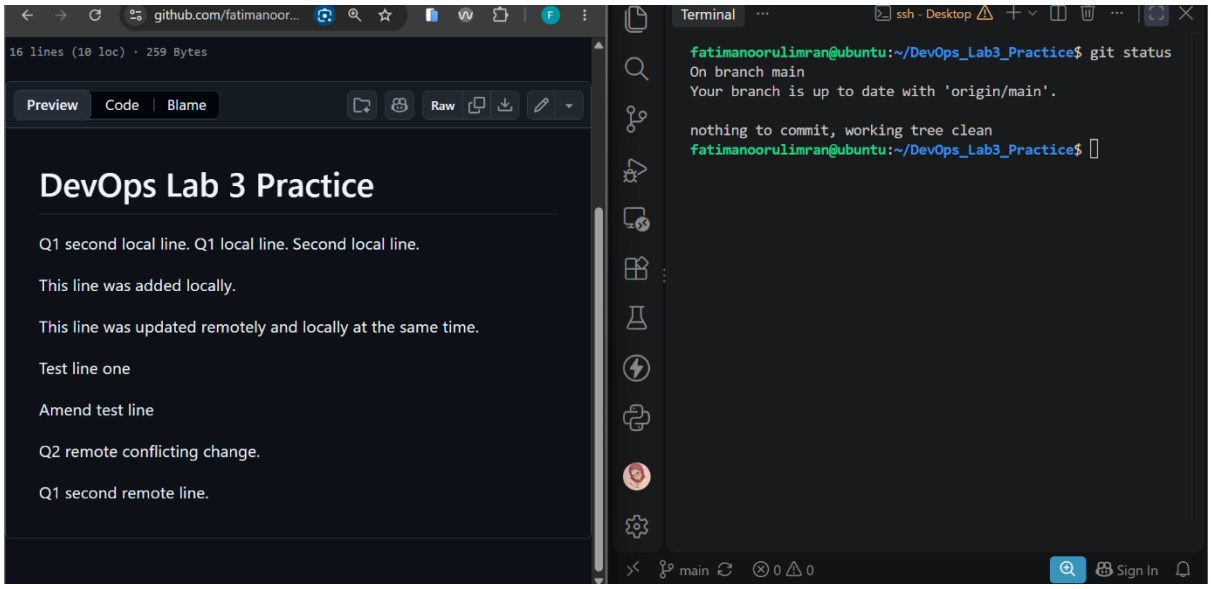
```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git pull --no-rebase origin main
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 2.05 KiB | 161.00 KiB/s, done.
From github.com:fatimanoorulimran/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   a1d4506..0527555  main       -> origin/main
Auto-merging README.md
Merge made by the 'ort' strategy.
 README.md | 13 +++----------
 1 file changed, 3 insertions(+), 10 deletions(-)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 770 bytes | 154.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   0527555..64e4f6a  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Resolved with git pull (merge) and pushed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1015 bytes | 169.00 KiB/s, done.
From github.com:fatimanoorulimran/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   64e4f6a..4b5d578  main       -> origin/main
Successfully rebased and updated refs/heads/main.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 379 bytes | 189.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   4b5d578..9cd4c6b  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Repeated with git pull --rebase and pushed.

Exam Question 2: Manual Merge Conflict Handling



Specific line changed on GitHub and committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ sed -i 's/Q1 remote line\./Q2 local conflicting change./' README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git diff
diff --git a/README.md b/README.md
index 4eac8cf..480e121 100644
--- a/README.md
+++ b/README.md
@@ -11,6 +11,6 @@ Test line one

 Amend test line

-Q1 remote line.
+Q2 local conflicting change.

 Q1 second remote line.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Q2 local conflicting change"
[main eea7d88] Q2 local conflicting change
 1 file changed, 1 insertion(+), 1 deletion(-)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Same line changed differently locally and committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:fatimanoorulimran/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Push rejected.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.00 KiB | 28.00 KiB/s, done.
From github.com:fatimanoorulimran/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   9cd4c6b..091a020  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply eea7d88... Q2 local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply eea7d88... # Q2 local conflicting change
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

git pull --rebase triggered the conflict.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps Lab 3 Practice
Q1 second local line.
Q1 local line.
Second local line.

This line was added locally.

This line was updated remotely and locally at the same time.

Test line one

Amend test line

Q2 remote and local conflicting change.

Q1 second remote line.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Conflict resolved manually in the file.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git rebase --continue
[detached HEAD 0935e26] Q2 local conflicting change
 1 file changed, 1 insertion(+), 1 deletion(-)
Successfully rebased and updated refs/heads/main.
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 378 bytes | 75.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   091a020..0935e26  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

File staged, rebase continued and changes pushed.

Exam Question 3: Managing Ignored and Tracked Files

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ mkdir DocFiles
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Document one" > DocFiles/doc1.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Document two" > DocFiles/doc2.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Document three" > DocFiles/doc3.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ ls DocFiles
doc1.txt  doc2.txt  doc3.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        DocFiles/

nothing added to commit but untracked files present (use "git add" to track)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

DocFiles folder created with several files.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add DocFiles
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Q3 add DocFiles folder"
[main 83eec5a] Q3 add DocFiles folder
 3 files changed, 3 insertions(+)
 create mode 100644 DocFiles/doc1.txt
 create mode 100644 DocFiles/doc2.txt
 create mode 100644 DocFiles/doc3.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 523 bytes | 174.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   0935e26..83eec5a  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Folder committed and pushed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "DocFiles/*" >> .gitignore
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ cat .gitignore
textfiles/*
DocFiles/*
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore

no changes added to commit (use "git add" and/or "git commit -a")
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

DocFiles added to .gitignore.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add .gitignore
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Q3 ignore DocFiles"
[main 156f225] Q3 ignore DocFiles
 1 file changed, 1 insertion(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 381 bytes | 127.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   83eec5a..156f225  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git ls-files DocFiles
DocFiles/doc1.txt
DocFiles/doc2.txt
DocFiles/doc3.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

.gitignore update committed and pushed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git rm -r --cached DocFiles
rm 'DocFiles/doc1.txt'
rm 'DocFiles/doc2.txt'
rm 'DocFiles/doc3.txt'
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        deleted:    DocFiles/doc1.txt
        deleted:    DocFiles/doc2.txt
        deleted:    DocFiles/doc3.txt

fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Folder removed from tracking with git rm -r --cached.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Q3 remove DocFiles from tracking"
[main 0b8f5a6] Q3 remove DocFiles from tracking
 3 files changed, 3 deletions(-)
 delete mode 100644 DocFiles/doc1.txt
 delete mode 100644 DocFiles/doc2.txt
 delete mode 100644 DocFiles/doc3.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 250 bytes | 250.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:fatimanoorulimran/DevOps_Lab3_Practice.git
   156f225..0b8f5a6  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git ls-files
.gitignore
README.md
notes.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ ls
DocFiles  README.md  notes.txt  textfiles
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Folder no longer present on GitHub.

Exam Question 4: Commit History Manipulation and Recovery

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Q4 line one" > q4.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add q4.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Q4 first commit"
[main 6c5950a] Q4 first commit
 1 file changed, 1 insertion(+)
 create mode 100644 q4.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

First change committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ echo "Q4 line two" >> q4.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git add q4.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Q4 second commit"
[main 287798a] Q4 second commit
 1 file changed, 1 insertion(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Second change committed.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
287798a (HEAD -> main) Q4 second commit
6c5950a Q4 first commit
0b8f5a6 (origin/main, origin/HEAD) Q3 remove DocFiles from tracking
156f225 Q3 ignore DocFiles
83eec5a Q3 add DocFiles folder
0935e26 Q2 local conflicting change
091a020 Q2 remote conflicting change
9cd4c6b Q1 second local edit
4b5d578 Q1 second remote edit
64e4f6a Merge branch 'main' of github.com:fatimanoorulimran/DevOps_Lab3_Practice
acfebb1 Q1 local edit
0527555 Q1 remote edit
39cf443 Q1 remote edit
a1d4506 Add body text
077a542 (origin/test-force, test-force) Revert "Added temporary text file"
233cc66 Added temporary text file
c6c86cb Fix log message and add notes
fb5905c Added test line
407a5fe Removed tracked textfiles directory
aad794b Added .gitignore to ignore textfiles directory
66d71f7 Added textfiles directory with three files
5bd44d3 Local conflicting change
306e660 Remote conflicting change
6c78846 Second local update
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Commit history shown.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git reset --soft HEAD~1
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
6c5950a (HEAD -> main) Q4 first commit
0b8f5a6 (origin/main, origin/HEAD) Q3 remove DocFiles from tracking
156f225 Q3 ignore DocFiles
83eec5a Q3 add DocFiles folder
0935e26 Q2 local conflicting change
091a020 Q2 remote conflicting change
9cd4c6b Q1 second local edit
4b5d578 Q1 second remote edit
64e4f6a Merge branch 'main' of github.com:fatimanoorulimran/DevOps_Lab3_Practice
acfebb1 Q1 local edit
0527555 Q1 remote edit
39cf443 Q1 remote edit
a1d4506 Add body text
```

Soft reset: history moved back, changes kept and staged.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Q4 third commit"
[main 4292946] Q4 third commit
 1 file changed, 1 insertion(+)
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -4
4292946 (HEAD -> main) Q4 third commit
6c5950a Q4 first commit
0b8f5a6 (origin/main, origin/HEAD) Q3 remove DocFiles from tracking
156f225 Q3 ignore DocFiles
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$
```

Changes committed again after the soft reset.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at 6c5950a Q4 first commit
fatimanoorulimran@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -4
6c5950a (HEAD -> main) Q4 first commit
0b8f5a6 (origin/main, origin/HEAD) Q3 remove DocFiles from tracking
156f225 Q3 ignore DocFiles
83eec5a Q3 add DocFiles folder
```

Hard reset: history moved back and the changes discarded.